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An Explainable and Trustworthy Multi-Agent Architecture for Job-Candidate Recommendation with Calibrated Confidence

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Harsh Kashyap: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Writing – review & editing.
Taranumpreet Kaur Wasu: Conceptualization, Software, Investigation, Data Curation, Writing – original draft, Writing – review & editing.

Parteek Kumar: Conceptualization, Supervision, Writing – review & editing, Funding acquisition.

Data availability

The frozen demo corpus (30 resumes, 15 job descriptions, 47 labeled pairs), the synthetic corpus and generator, the evaluation scripts, the explanation generator, the calibration layer, and the regression benchmark will be deposited in a public repository (Harvard Dataverse) with a citable DOI upon acceptance; a fully anonymized copy is available to reviewers during review. The release includes the one-command reproduction script, the auto-generated tables, the per-experiment JSON artifacts, and the figure source scripts.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used OpenAI ChatGPT (GPT-4 model) as a language-model assistant to support copy-editing of selected paragraphs, to generate initial drafts of figure captions from a structured outline, and to suggest edits during the literature review. All technical content, experimental results, design decisions, and final wording are the authors' own. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.